

1. Queries & Descriptions

SQL Query 1 (Complex)

Query: Return ZIP codes in a user-specified city or state, along with average housing price, total income, number of schools, and an income-to-price ratio.

Description: This query supports the main search results page by combining housing, income, and education data into one ZIP-level summary. It helps users compare neighborhoods within a selected city or state.

```
WITH IncomeSummary AS (  
    SELECT  
        zip_code,  
        SUM(num_returns) AS total_returns,  
        SUM(total_income) AS total_income  
    FROM IRS_Income  
    GROUP BY zip_code  
)  
,  
SchoolSummary AS (  
    SELECT  
        zip_code,  
        COUNT(*) AS num_schools,  
        AVG(enrollment) AS avg_school_enrollment  
    FROM Education  
    GROUP BY zip_code  
)  
,  
HousingSummary AS (  
    SELECT  
        zip_code,  
        AVG(price) AS avg_housing_price,  
        AVG(house_size) AS avg_house_size  
    FROM Real_Estate  
    WHERE price IS NOT NULL  
        AND price >= 50000  
    GROUP BY zip_code  
)  
SELECT  
    l.zip_code,  
    l.city,  
    l.state,  
    h.avg_housing_price,  
    i.total_income,  
    COALESCE(s.num_schools, 0) AS num_schools,  
    COALESCE(s.avg_school_enrollment, 0) AS avg_school_enrollment,  
    i.total_income::numeric / NULLIF(h.avg_housing_price, 0) AS income_price_ratio
```

```

FROM Location l
JOIN HousingSummary h
    ON l.zip_code = h.zip_code
JOIN IncomeSummary i
    ON l.zip_code = i.zip_code
LEFT JOIN SchoolSummary s
    ON l.zip_code = s.zip_code
WHERE l.city = 'Philadelphia'
    OR l.state = 'PA'
ORDER BY income_price_ratio DESC, h.avg_housing_price ASC;

```

SQL Query 2 (Complex)

Query: Return ZIP codes where average housing price is below the state average, total income is above the state average, and the ZIP code has at least 3 schools.

Description: This query identifies ZIP codes that may offer strong value within their state by comparing each area against state-level benchmarks in housing and income.

```

WITH IncomeSummary AS (
    SELECT
        zip_code,
        SUM(total_income) AS total_income
    FROM IRS_Income
    GROUP BY zip_code
),
SchoolSummary AS (
    SELECT
        zip_code,
        COUNT(*) AS num_schools
    FROM Education
    GROUP BY zip_code
),
HousingSummary AS (
    SELECT
        zip_code,
        AVG(price) AS avg_housing_price
    FROM Real_Estate
    WHERE price IS NOT NULL
        AND price >= 50000
    GROUP BY zip_code
),
ZipSummary AS (
    SELECT
        l.zip_code,

```

```

        l.city,
        l.state,
        h.avg_housing_price,
        i.total_income,
        COALESCE(s.num_schools, 0) AS num_schools
    FROM Location l
    JOIN HousingSummary h ON l.zip_code = h.zip_code
    JOIN IncomeSummary i ON l.zip_code = i.zip_code
    LEFT JOIN SchoolSummary s ON l.zip_code = s.zip_code
),
StateBenchmarks AS (
    SELECT
        state,
        AVG(avg_housing_price) AS state_avg_housing_price,
        AVG(total_income) AS state_avg_total_income
    FROM ZipSummary
    GROUP BY state
)
SELECT
    z.zip_code,
    z.city,
    z.state,
    z.avg_housing_price,
    z.total_income,
    z.num_schools
FROM ZipSummary z
JOIN StateBenchmarks s
    ON z.state = s.state
WHERE z.avg_housing_price < s.state_avg_housing_price
    AND z.total_income > s.state_avg_total_income
    AND z.num_schools >= 3
ORDER BY z.total_income DESC, z.avg_housing_price ASC;

```

SQL Query 3 (Complex)

Query: Return ZIP codes where average housing price, total income, and number of schools all fall within user-defined ranges.

Description: This query acts as a multi-condition search engine for neighborhoods and supports combined filtering similar to an Expedia-style search interface.

```

WITH IncomeSummary AS (
    SELECT
        zip_code,
        SUM(total_income) AS total_income

```

```

FROM IRS_Income
GROUP BY zip_code
),
SchoolSummary AS (
    SELECT
        zip_code,
        COUNT(*) AS num_schools
    FROM Education
    GROUP BY zip_code
),
HousingSummary AS (
    SELECT
        zip_code,
        AVG(price) AS avg_housing_price
    FROM Real_Estate
    WHERE price IS NOT NULL
    AND price >= 50000
    GROUP BY zip_code
)
SELECT
    l.zip_code,
    l.city,
    l.state,
    h.avg_housing_price,
    i.total_income,
    COALESCE(s.num_schools, 0) AS num_schools
FROM Location l
JOIN HousingSummary h
    ON l.zip_code = h.zip_code
JOIN IncomeSummary i
    ON l.zip_code = i.zip_code
LEFT JOIN SchoolSummary s
    ON l.zip_code = s.zip_code
WHERE h.avg_housing_price BETWEEN 150000 AND 500000
    AND i.total_income BETWEEN 100000 AND 400000
    AND COALESCE(s.num_schools, 0) BETWEEN 3 AND 20
ORDER BY h.avg_housing_price ASC;

```

SQL Query 4 (Complex)

Query: Return ZIP codes where average price per square foot is below the national average, but total income is above the national average.

Description: This query helps identify neighborhoods that may offer strong housing value in relatively high-income areas.

```

WITH IncomeSummary AS (
    SELECT
        zip_code,
        SUM(total_income) AS total_income
    FROM IRS_Income
    GROUP BY zip_code
),
ZipPricePerSqft AS (
    SELECT
        zip_code,
        AVG(price::numeric / NULLIF(house_size, 0)) AS avg_price_per_sqft
    FROM Real_Estate
    WHERE house_size > 0
        AND price IS NOT NULL
        AND price >= 50000
    GROUP BY zip_code
),
NationalBenchmarks AS (
    SELECT
        AVG(price::numeric / NULLIF(house_size, 0)) AS national_avg_price_per_sqft
    FROM Real_Estate
    WHERE house_size > 0
        AND price IS NOT NULL
        AND price >= 50000
),
NationalIncome AS (
    SELECT
        AVG(total_income) AS national_avg_total_income
    FROM (
        SELECT zip_code, SUM(total_income) AS total_income
        FROM IRS_Income
        GROUP BY zip_code
    ) t
)
SELECT
    l.zip_code,
    l.city,
    l.state,
    z.avg_price_per_sqft,
    i.total_income
FROM Location l
JOIN ZipPricePerSqft z
    ON l.zip_code = z.zip_code

```

```

JOIN IncomeSummary i
  ON l.zip_code = i.zip_code
CROSS JOIN NationalBenchmarks nb
CROSS JOIN NationalIncome ni
WHERE z.avg_price_per_sqft < nb.national_avg_price_per_sqft
  AND i.total_income > ni.national_avg_total_income
ORDER BY i.total_income DESC, z.avg_price_per_sqft ASC;

```

SQL Query 5

Query: Return ZIP codes sorted by highest total income.

Description: This query provides a simple ranking of neighborhoods by total reported income.

```

WITH IncomeSummary AS (
  SELECT
    zip_code,
    SUM(total_income) AS total_income
  FROM IRS_Income
  GROUP BY zip_code
)
SELECT
  l.zip_code,
  l.city,
  l.state,
  i.total_income
FROM Location l
JOIN IncomeSummary i
  ON l.zip_code = i.zip_code
ORDER BY i.total_income DESC;

```

SQL Query 6

Query: Return ZIP codes sorted by lowest average housing price.

Description: This query helps users explore more affordable neighborhoods. Listings below a minimum price threshold are excluded to avoid unrealistic results.

```

SELECT
  l.zip_code,
  l.city,
  l.state,
  AVG(r.price) AS avg_housing_price
FROM Location l
JOIN Real_Estate r
  ON l.zip_code = r.zip_code

```

```
WHERE r.price IS NOT NULL
      AND r.price >= 50000
GROUP BY l.zip_code, l.city, l.state
ORDER BY avg_housing_price ASC;
```

SQL Query 7

Query: For a selected ZIP code, return average housing price, average house size, average number of bedrooms, and average number of bathrooms.

Description: This query summarizes the local housing market for the selected ZIP code.

```
SELECT
  zip_code,
  AVG(price) AS avg_housing_price,
  AVG(house_size) AS avg_house_size,
  AVG(bedrooms) AS avg_bedrooms,
  AVG(bathrooms) AS avg_bathrooms
FROM Real_Estate
WHERE zip_code = '19104'
      AND price IS NOT NULL
      AND price >= 50000
GROUP BY zip_code;
```

SQL Query 8

Query: For a selected ZIP code, return the total number of schools and the average school enrollment.

Description: This query summarizes education-related characteristics for the selected ZIP code.

```
SELECT
  zip_code,
  COUNT(*) AS total_schools,
  AVG(enrollment) AS avg_school_enrollment
FROM Education
WHERE zip_code = '19104'
GROUP BY zip_code;
```

SQL Query 9

Query: For a selected ZIP code, return total income, wage income, interest income, dividend income, and capital gains.

Description: This query provides a ZIP-level economic summary using the IRS dataset.

```

SELECT
    zip_code,
    SUM(total_income) AS total_income,
    SUM(wage_income) AS total_wage_income,
    SUM(interest_income) AS total_interest_income,
    SUM(dividend_income) AS total_dividend_income,
    SUM(capital_gain) AS total_capital_gain
FROM IRS_Income
WHERE zip_code = '19104'
GROUP BY zip_code;

```

SQL Query 10

Query: For a selected ZIP code, return the income statistics broken down by income bracket.

Description: This query shows how income-related measures vary across brackets within the selected ZIP code.

```

SELECT
    zip_code,
    income_bracket,
    num_returns,
    total_income,
    wage_income,
    interest_income,
    dividend_income,
    capital_gain
FROM IRS_Income
WHERE zip_code = '19104'
ORDER BY income_bracket;

```

2. Credentials

Database Platform: PostgreSQL (AWS RDS)

Host: database-1.cijymfgets4w.us-east-1.rds.amazonaws.com

Port: 5432

Username: FinalProject5500

Password: hjzsjjqhzbm

Database: housing_app

Instructions:

The database can be accessed using any PostgreSQL client such as DataGrip or psql. In DataGrip, create a new PostgreSQL connection and enter the host, port, database, username, and password above.

3. Github link

<https://github.com/bobbychia/CIS-5500-FinalProject-Group22>